

Machine	Temp	Vibration	Failure
1	H	H	Yes
2	H	L	Yes
3	M	H	Yes
4	L	L	No
5	L	M	No
6	M	L	No
7	H	M	Yes
8	L	H	No

Total Entropy = (2) 1/2

Variable $\rightarrow$ Failure

Yes $\rightarrow 4$

No $\rightarrow 4$

Total = 8

$$\text{Entropy} = -P_1 \log_2 P_1 - P_2 \log_2 P_2$$

$$= -\frac{4}{8} \log_2 \left(\frac{4}{8}\right) - \frac{4}{8} \log_2 \left(\frac{4}{8}\right)$$

$$= -0.5(-1) - 0.5(-1)$$

$$= 1$$

1. Information gain

Temp' (how)

Yes = 0

No = 3

$$\text{Entropy} = \log_2 \frac{3}{3} = 0$$

Temp. (Med)

Yes = 1

No = 1

$$\text{Entropy} = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1$$

Temp (Agh)

Yes = 3

No = 0

$$\text{Entropy} = \log_2 \frac{3}{3} = 0$$

Weighted Entropy of temp

$$= \frac{3}{8}(0) + \frac{2}{8}(1) + \frac{3}{8}(0) = \underline{0.25}$$

Information gain (I.G)

$$\text{I.G} = \text{Entropy} - \text{Weighted Entropy}$$

$$= 1 - 0.25 = \underline{0.75}$$

2. Information gain of Vibration

Vib (High)

Yes = 2

No = 1

$$\text{Entropy} = \frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} \\ = \underline{\underline{0.918}}$$

Vib (Med)

Yes = 1

No = 1

$$\text{Entropy} = 1$$

Vib (Low)

Yes = 1

No = 2

$$\text{Entropy} = \frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} \\ = \underline{\underline{0.918}}$$

Weighted Entropy:

$$W(E) = \frac{3}{8} (0.918) + \frac{1}{8} (1) + \frac{3}{8} (0.918) \\ = 0.938$$

$$I.C. = E - W(E)$$

$$= 1 - 0.938 = \underline{\underline{0.062}}$$

Comparison

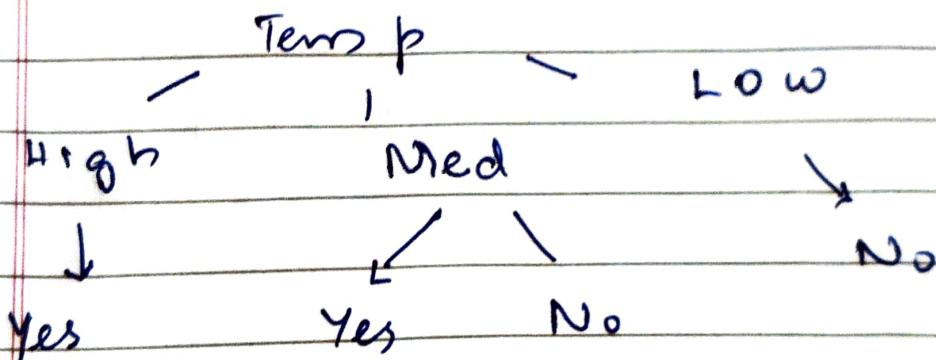
$$I.C (T) = 0.75$$

$$I.C (V) = 0.062$$

$$0.75 > 0.062$$

$\therefore$ Temp has priority

Decision tree



For temp = high & Vib = Med

